Embeddings as a Bridge Between Neural and Classical Models: A Systematic Study of Representation Transfer in Tabular Recommendation Systems

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Abstract

Background. Learned dense embeddings for categorical features have demonstrated strong performance in neural recommendation models. Their utility when transferred as pre-computed feature representations to classical machine learning models remains systematically underexplored. **Methods.** We conduct a controlled three-condition ablation study across three publicly available recommendation datasets (a 100K temporally sampled subset of MovieLens 20M, Amazon Product Reviews, and Yelp Academic Dataset), four algorithms (Linear Regression, Random Forest, XGBoost, and a feedforward ANN), and three experimental conditions: OHE baseline with default hyperparameters; ANN-learned embeddings applied to all models with fixed hyperparameters; and Optuna TPE hyperparameter optimisation on embedding features. **Results.** Embedding transfer benefit is heterogeneous across model classes. Linear Regression showed consistently small improvements across all datasets (maximum delta: 0.0146 RMSE), consistent with its limited capacity to exploit the relational structure of embedding space. Tree models benefited only when combined with hyperparameter tuning. The ANN showed the largest improvements, scaling with baseline overfitting severity: negligible on MovieLens (+0.0005), meaningful on Amazon (+0.0775), and substantial on Yelp (+0.3299). XGBoost achieved best performance on the two larger datasets (MovieLens: 0.9274 RMSE; Amazon: 1.3579 RMSE), while Linear Regression performed best on the smallest dataset (Yelp: 1.5526 RMSE). **Conclusions.** ANN-learned embeddings transfer effectively to classical models, but the magnitude of benefit depends on (1) the receiving model's capacity to act on relational structure in embedding space, and (2) how severely the alternative encoding was causing overfitting. For the ANN specifically, the transfer condition reflects representation substitution rather than cross-architecture transfer. Target leakage via aggregated interaction features represents a recurring methodological risk requiring explicit Leave-One-Out mitigation.

Keywords: recommender systems, representation learning, categorical embeddings, transfer learning, gradient boosting, tabular learning, feature engineering

1. Introduction

The encoding of categorical features is a foundational decision in recommendation system design. One-hot encoding (OHE) remains prevalent in classical machine learning pipelines due to its simplicity. Learned dense embeddings, trained end-to-end in neural architectures, position categories in continuous vector spaces where relational structure emerges from training signal [1,2].

A practical question arises from this: can the representations learned by a neural embedding model be extracted and reused as feature columns by classical models? Consider the concrete case: an XGBoost model that has never seen an embedding layer receives pre-computed embedding vectors as input features. Does it benefit?

We call this the *embedding transfer problem*. It has direct practical relevance for practitioners who wish to leverage neural representation learning without committing fully to end-to-end neural architectures. We address three research questions:

- **RQ1:** Do ANN-learned categorical embeddings improve classical recommendation models when transferred as pre-computed feature representations?
- **RQ2:** What determines the magnitude of transfer benefit across model classes?
- **RQ3:** Is hyperparameter optimisation a necessary complement to embedding transfer for classical models to realise benefit?

We address these questions through a controlled three-condition ablation study across three datasets with deliberately contrasting characteristics.

2. Related Work

Matrix factorisation methods [3] established latent factor representations for collaborative filtering. Neural Collaborative Filtering [4] extended this to non-linear interaction modelling, demonstrating advantages of jointly optimised embeddings and network weights over separately trained components.

Guo and Berkhahn [5] formalised entity embeddings for categorical variables in tabular learning, showing that neural networks learn meaningful dense representations for non-NLP categorical features. Subsequent work [6,7] explored transfer of such representations to gradient boosted trees with mixed results. Our study contextualises these findings across a broader and more controlled experimental framework.

Grinsztajn et al. [9] demonstrated consistent tree-based model superiority on tabular data. Our results partially replicate this finding and qualify it with a dataset size condition: on the smallest dataset, Linear Regression outperforms tree methods under optimal conditions.

3. Methodology

3.1 Experimental Design

We use a three-condition within-dataset design to separate the contributions of encoding strategy and hyperparameter optimisation:

Condition 1 (Baseline). Standard OHE for categorical variables and multi-hot encoding for multi-label features. All models use default scikit-learn and XGBoost hyperparameters.

Condition 2 (Transfer). An EmbeddingANN is trained end-to-end, learning dense representations for all categorical features. Embedding weight matrices are extracted post-training and concatenated with scaled numerical features for classical model training. Hyperparameters remain at default values, isolating the encoding effect from the tuning effect.

Condition 3 (Optimised). Optuna TPE sampler [10] performs hyperparameter search on the embedding feature matrices. Linear Regression uses RidgeCV (analytical alpha sweep). Random Forest: 30 trials. XGBoost: 50 trials with early stopping. ANN: 50 trials including architecture depth search (1 to 5 layers).

An important conceptual distinction applies to the ANN in Condition 2. For Linear Regression, Random Forest, and XGBoost, this condition constitutes genuine cross-architecture transfer: pre-computed embedding weights produced by one model architecture are consumed as static features by a different model class. For the ANN, Condition 2 represents representation substitution rather than transfer. The ANN already learns embeddings internally during training. The difference between Conditions 1 and 2 for the ANN is the replacement of sparse OHE inputs with learned dense inputs, not a change in the source architecture. Benefit for the ANN therefore reflects embedding compactness and information density, not cross-model transfer.

3.2 Datasets

Dataset	Sample	Rating scale	Categoricals	Split
MovieLens 20M (100K sample)	100K	0.5-5.0	Genres, Language	Fixed calendar dates
Amazon Reviews	500K	1-5 integer	userId, productId	Percentile 70/15/15
Yelp Academic	100K	1-5 integer	City, State, Categories	Percentile 70/15/15

Table 1: Dataset summary. MovieLens uses a 100K temporally sampled subset of the full 20M rating dataset.

All splits are temporal. Training data precedes validation and test data chronologically. Random splits were explicitly rejected to prevent future interaction leakage into training.

3.3 Feature Engineering and Target Leakage Prevention

For Amazon and Yelp, interaction-derived features require Leave-One-Out (LOO) aggregation to prevent target leakage. For a training row with user u and rating r_i:

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user_avg(u, i) = (sum_of_all_ratings_by_u - r_i) / (count_by_u - 1)
```

This ensures the feature for row i does not contain r_i. The naive alternative (computing user averages from the full training set and merging back onto training rows) produced Train RMSE of 0.04 and Test RMSE of 1.70 on Amazon. A gap of 1.66 confirms the scale of leakage when this precaution is omitted.

Cold-start entities receive global average fallbacks computed from the per-entity statistics table, not the row-expanded training frame. Active entities must not receive disproportionate weight in the global average.

3.4 EmbeddingANN Architecture

The EmbeddingANN processes each categorical feature through a dedicated nn.Embedding table. Multi-label features (genres, business categories) use masked mean-pooling across token embeddings to produce fixed-size representations regardless of how many tokens a row contains.

High-cardinality ID features (userId, productId, city) apply vocabulary filtering: only entities with 5 or more training interactions receive dedicated embedding vectors. Entities below this threshold receive a zero vector and rely entirely on numerical features. This prevents embedding table memorisation, which occurred when unrestricted vocabularies produced more parameters than training rows.

During Condition 3 optimisation, Optuna tunes the number of hidden layers (1 to 5), per-feature embedding dimensions, dropout rates, learning rate, and weight decay. Early stopping with patience of 5 epochs is applied throughout.

3.5 Evaluation Protocol

Primary metric: Root Mean Square Error (RMSE) on the temporal test split. Secondary: Mean Absolute Error (MAE). Train-Test RMSE gap serves as an overfitting diagnostic throughout. All seeds fixed at 42. Multi-seed variance analysis is identified as a limitation of this study.

4. Results

4.1 MovieLens 20M (100K Temporally Sampled Subset)

Model	Baseline	Transfer	Optimised	Emb Change	Opt Change
Linear Regression	0.9381	0.9293	0.9293	-0.0088	0.0000
Random Forest	0.9641	0.9340	0.9279	-0.0301	-0.0061
XGBoost	0.9394	0.9319	0.9274+	-0.0075	-0.0045
ANN	0.9338	0.9333	0.9300	-0.0005	-0.0033

Table 2: MovieLens Test RMSE. + = best overall. Emb Change = Transfer minus Baseline. Opt Change = Optimised minus Transfer.

Performance improvements on MovieLens are uniformly small, with a total range of 0.0038 to 0.0362 RMSE across all conditions. Embedding substitution provided larger improvements than optimisation for most models. The ANN showed negligible embedding benefit (+0.0005), consistent with the hypothesis that benefit scales with baseline overfitting severity: the MovieLens OHE matrix has approximately 50 columns and causes no meaningful memorisation. The feature ceiling (rich IMDB and TMDB metadata already present) limits what any encoding or tuning change can add.

4.2 Amazon Product Reviews

Model	Baseline	Transfer	Optimised	Emb Change	Opt Change
Linear Regression	1.3646	1.3645	1.3644	-0.0001	-0.0001
Random Forest	1.5559	1.5153	1.3626	-0.0406	-0.1527
XGBoost	1.4942	1.4764	1.3579+	-0.0178	-0.1185
ANN	1.4746	1.3971	1.3729	-0.0775	-0.0242

Table 3: Amazon Test RMSE. + = best overall.

Amazon results show a qualitatively different pattern from MovieLens. The total improvement range is 0.1933, an order of magnitude larger. For tree models, optimisation provided substantially larger improvements than embedding transfer (Random Forest: 0.1527 vs 0.0406; XGBoost: 0.1185 vs 0.0178). The default max_depth=None setting in Random Forest was the primary bottleneck, producing a Train-Test gap of 0.976 at baseline.

ANN embedding benefit (+0.0775) was meaningfully larger than LR benefit (+0.0001). The initial Amazon EmbeddingANN training produced validation RMSE that increased from 1.86 to 7.50 over 20 epochs, caused by embedding table memorisation (400,000+ unique users at 32 dimensions yields 13 million parameters against 350,000 training rows). Vocabulary filtering (minimum 5 interactions), reduced embedding dimensions (8), dropout (0.5), and early stopping resolved this.

4.3 Yelp Academic Dataset

Model	Baseline	Transfer	Optimised	Emb Change	Opt Change
Linear Regression	1.5672	1.5526+	1.5542	-0.0146	+0.0016
Random Forest*	1.5831	1.7030	1.5820	+0.1199	-0.1210
XGBoost	1.6517	1.6716	1.5597	+0.0199	-0.1119
ANN	1.9168	1.5869	1.5871	-0.3299	+0.0002

Table 4: Yelp Test RMSE. + = best overall. * RF baseline used constrained hyperparameters (max_depth=15, max_features=sqrt, max_samples=0.5) for computational feasibility with the approximately 1,944-column sparse OHE baseline matrix.

Three notable findings arise from the Yelp results.

First, the ANN embedding substitution benefit (+0.3299) is the largest in the study. The baseline OHE matrix (1,944 columns, 70,000 training rows) produced a Train-Test gap of 1.133. Replacing it with 58 dense embedding features reduced the gap to 0.295. This is the clearest demonstration of the representation substitution mechanism.

Second, embedding transfer degraded tree model performance before optimisation. Random Forest increased from 1.5831 to 1.7030 under Condition 2, and XGBoost from 1.6517 to 1.6716. Dense features provide more information per split, enabling deeper memorisation when depth is unconstrained. This confirms that embedding transfer to tree models requires accompanying regularisation to be beneficial.

Third, Linear Regression achieves the best overall result (1.5526) from Condition 2. On 70,000 training rows, the simplest regularised model generalises most effectively. This constitutes a dataset size effect on optimal model complexity that holds across all three datasets.

5. Discussion

5.1 Determinants of Transfer Benefit

Our results identify two primary determinants of embedding transfer benefit.

The first is receiving model capacity to act on relational structure. Embedding space positions similar entities in proximity. An Italian restaurant embedding clusters near French restaurant embeddings because users who rate one similarly tend to rate the other. A high-volume reviewer's user embedding clusters near other high-volume reviewers. Linear Regression can access embedding coordinates directly, but has limited capacity to exploit cluster membership or inter-point distances. Acting on proximity requires non-linear decision boundaries. This accounts for the consistent near-zero LR improvement across all three datasets. Tree models and neural networks can more fully exploit this relational structure, but tree models require proper regularisation to avoid memorising it.

The second determinant is the severity of baseline encoding overfitting. ANN benefit scales directly with how badly the OHE baseline caused memorisation. On MovieLens (manageable OHE matrix), the benefit was +0.0005. On Yelp (1,944-column sparse matrix), it was +0.3299. This suggests embedding transfer is most valuable precisely when the alternative encoding is most problematic.

5.2 ANN Transfer vs Representation Substitution

A consistent but underemphasised distinction in the embedding transfer literature is whether the downstream model is the same architecture that produced the embeddings. In our study, for Linear Regression, Random Forest, and XGBoost, Condition 2 constitutes genuine cross-architecture transfer: a feedforward neural network produces embedding weights that are consumed as static features by fundamentally different model classes.

For the ANN, Condition 2 is different. The ANN learns embeddings internally during training regardless of input format. The distinction between Conditions 1 and 2 for the ANN is that sparse OHE inputs are replaced with learned dense inputs. The benefit measured for the ANN is therefore the value of dense versus sparse categorical representations in the same architecture, not the value of cross-model knowledge transfer.

5.3 Dataset Size and Optimal Model Complexity

XGBoost achieved best performance under optimal conditions on MovieLens (100K rows, rich metadata) and Amazon (500K rows). Linear Regression achieved best performance on Yelp (100K rows, smaller effective training set of 70K). This is consistent with the bias-variance tradeoff: on smaller datasets, variance from complex models outweighs their lower bias, and simpler well-regularised models generalise more reliably [9].

5.4 Methodological Contributions

Two methodological findings have broader applicability. First, the target leakage scenario: computing user or item statistics from the full training set and merging back onto training rows inflates training accuracy while destroying generalisation. Leave-One-Out aggregation is a necessary precaution, not an optional refinement.

Second, the relative importance of encoding strategy versus hyperparameter optimisation is not universal. On MovieLens, embedding substitution contributed more than tuning for most models. On Amazon, optimisation contributed substantially more than encoding change for tree models. Studies comparing encoding strategies at fixed (often default) hyperparameters may attribute to encoding what is actually attributable to overfitting.

6. Limitations

- All results are single-run with seed 42. Multi-seed variance analysis is required before strong statistical claims can be made.
- No explicit collaborative filtering baselines (SVD, ALS, NMF) are included.
- Temporal rolling features (user activity windows) use a fixed reference date rather than per-row timestamps, introducing minor forward leakage within the training set. Validation and test sets are unaffected.
- Yelp Random Forest baseline used constrained hyperparameters for computational feasibility with the wide sparse feature matrix. This is noted in all results tables.

7. Conclusion

We presented a controlled three-condition ablation study of categorical embedding transfer across three recommendation datasets and four algorithm classes. The central finding is that ANN-learned embeddings transfer to classical models, with benefit determined by two factors: the receiving model's capacity to exploit relational structure in embedding space, and the severity of overfitting caused by the baseline encoding strategy.

For the ANN specifically, the transfer condition measures representation substitution (dense vs sparse inputs in the same architecture) rather than cross-model transfer. This distinction matters for interpreting ANN results in embedding transfer studies.

Key empirical findings: LR benefit was consistently small; tree model benefit required accompanying regularisation; ANN benefit scaled with baseline overfitting severity; XGBoost performed best on larger datasets while LR performed best on the smallest; and target leakage via aggregated features is a recurring methodological risk.

Future work should incorporate sequence-aware architectures, explicit collaborative filtering baselines, and multi-seed variance analysis.

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